College Physics 2

THERMAL PHYSICS

MPS Department I FEU Institute of Technology

College Physics 2

THERMAL EXPANSION

MPS Department I FEU Institute of Technology

OBJECTIVES

- Explain the basic concepts of thermal physics
- Define temperature and thermal expansion
- Solve problems related to thermal expansion

DEFINITION OF TERMS

Temperature – it is the degree of hotness of an object and it can be measured by thermometer.

Usually expressed in **Celsius**, **Fahrenheit**, or **Kelvin** Scale.

Heat – it is the energy which travels from high temperature to low temperature in a matter. It is also called **thermal or internal energy.**

DEFINITION OF TERMS

Thermal Physics is the branch of physics that deals with the study of temperature, heat energy and their relation to matter.

When objects are heated or cooled, the temperature changes and some properties of the materials change. This is called **Thermometric Properties.**

Thermometer

Mercury is a common liquid element used in thermometer. It freezes at -39° C and boils at 357° C.

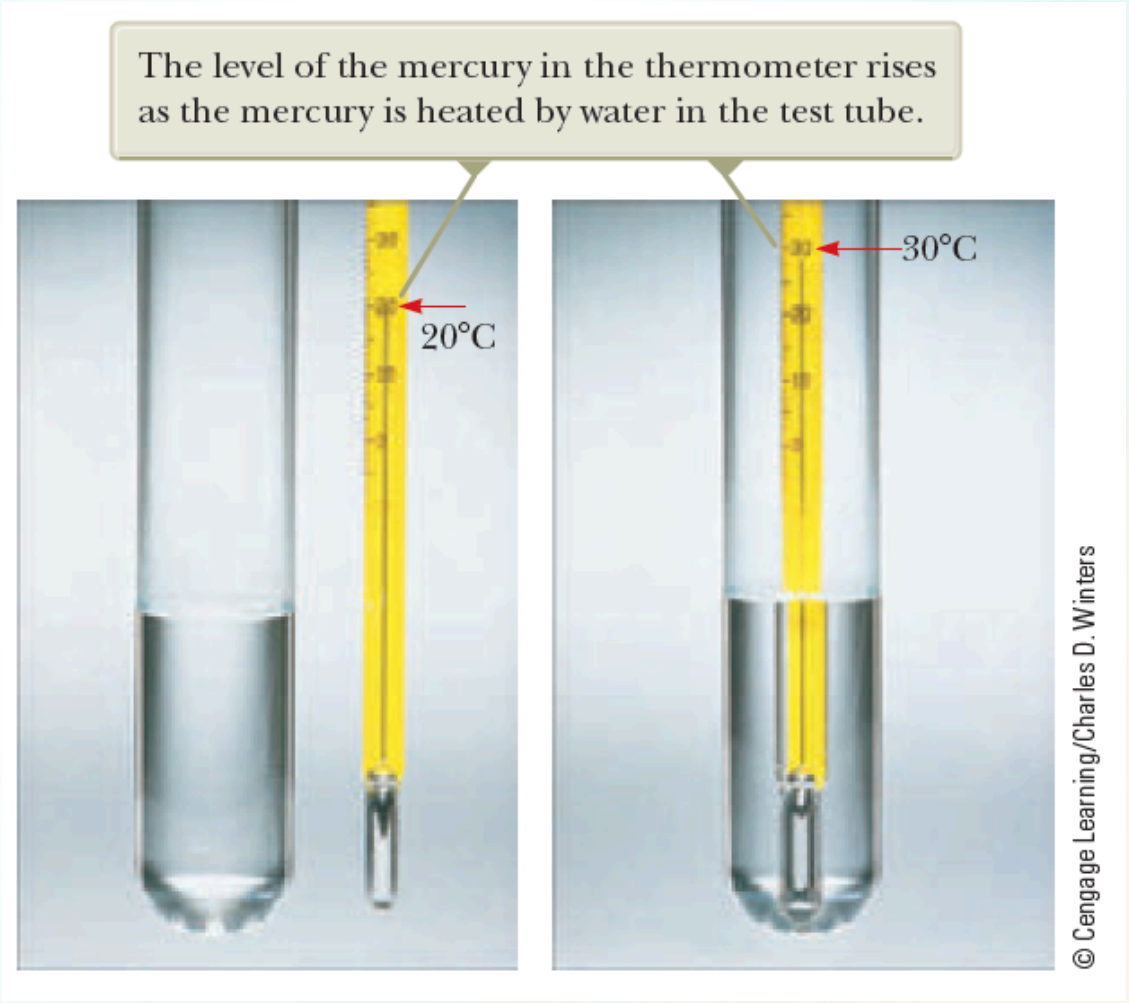


Figure 1. A mercury thermometer before and after increasing its temperature. From. *Physics for Scientists and Engineers with Modern Physics*(pp 570) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

CONVERSION

Conversion of Celsius to Fahrenheit:

$$T_F = \frac{9}{5}T_C + 32°$$

Conversion of Farenheit to Celsius:

$$T_C = \frac{5}{9}(T_F - 32°)$$

EXPANSION OF SOLIDS , LIQUIDS and GASES

All matters expand when heated because of the increase in the vibration of the molecules.

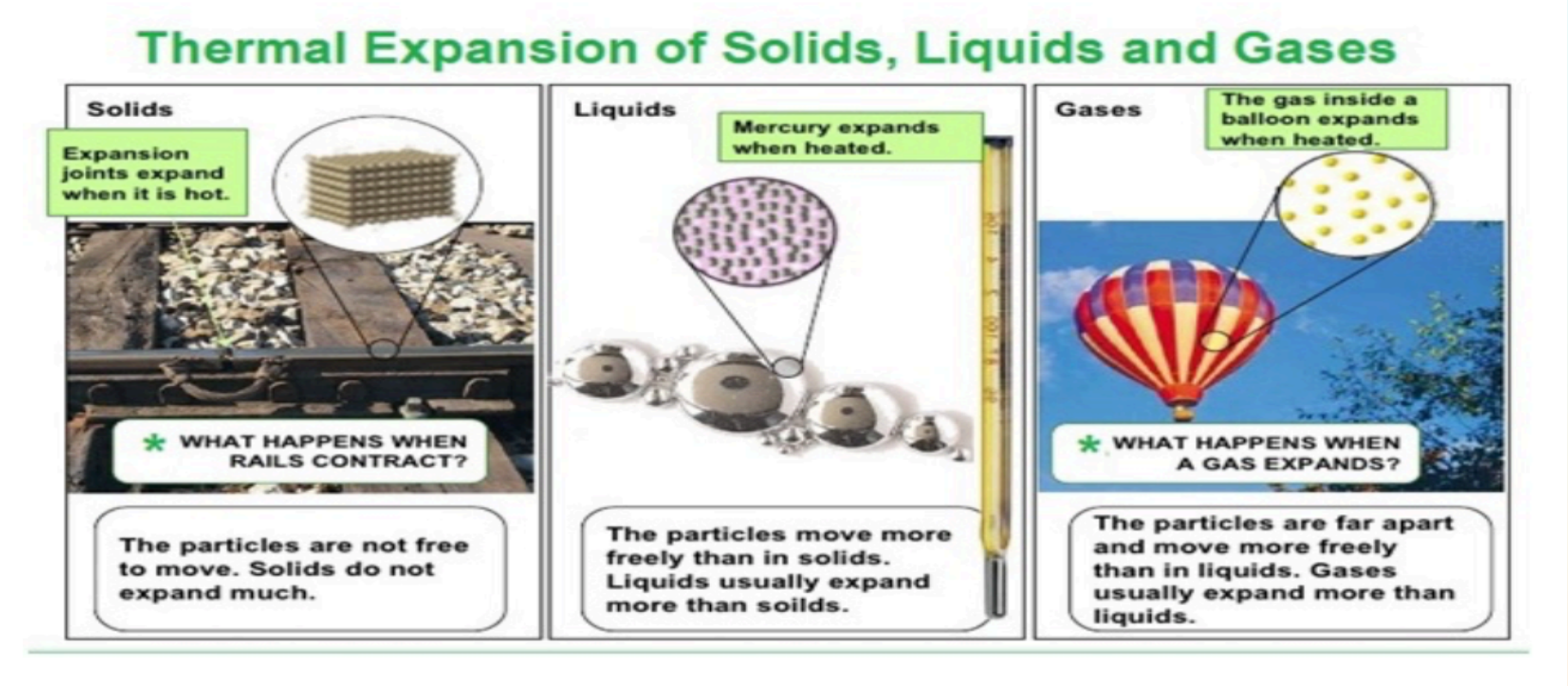


*Figure 2 Ther*mal Expansion of Solids, Liquids and Gases. From. https://www.jagranjosh.com/general-knowledge/thermal-expansion-of-solids-liquids-and-gases-1458720400-1

LINEAR EXPANSION

A rod of a material has a length L_o at some initial temperature T_0. When the temperature changes by ΔT, the length changes by ΔL.

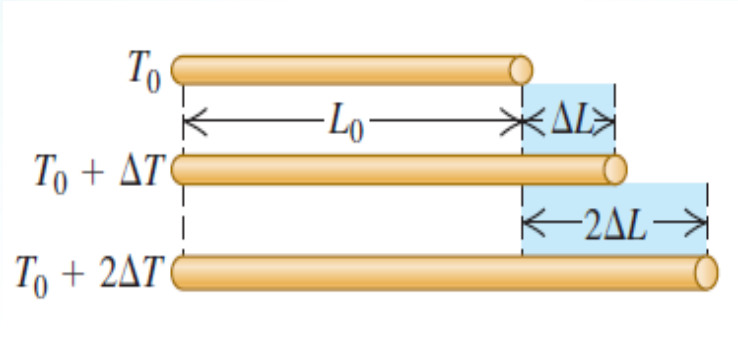


Figure 3. How the length of a rod changes with a change in temperature. (Length changes are exaggerated for clarity.)From. *Sear's & Zemansky's University Physics With Modern Physics*(pp557) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

For moderate temperature changes, change in length is directly proportional to change in temperature.

LINEAR EXPANSION

The formula for linear expansion is given as follows:

$$\Delta L = \alpha L_o \Delta T$$

If a rod has length L_o at temperature T_o, then its length L at a temperature $T = T_o + \Delta T$ is,

$$L = L_o + \Delta L = L_o + \alpha L_o \Delta T$$

$$\boldsymbol{L = L_o(1 + \alpha \Delta T)}$$

Where: ΔL – change in length (m)
L_0– Initial length (m)
L – final length (m)
ΔT – change in temperature (C^0)
α – coefficient of linear expansion ($/C^0$)

SAMPLE

1. A Surveyors uses a steel measuring tape is exactly 50 m long at a temperature of 20°C. The markings on the tape are calibrated for this temperature. What is the length of the tape when the temperature is 35°C? (α_{steel} = 11x 10^{-6} /C^0)

SOLUTION

$$L = L_o(1 + \alpha\Delta T)$$

= 50m [1+ (11x 10^{-6} /C^0) (35-20)C^0]

= 50.008m

ANSWER: 50.008m

AREA EXPANSION

When an object undergoes thermal expansion, any holes in the object expand as well.

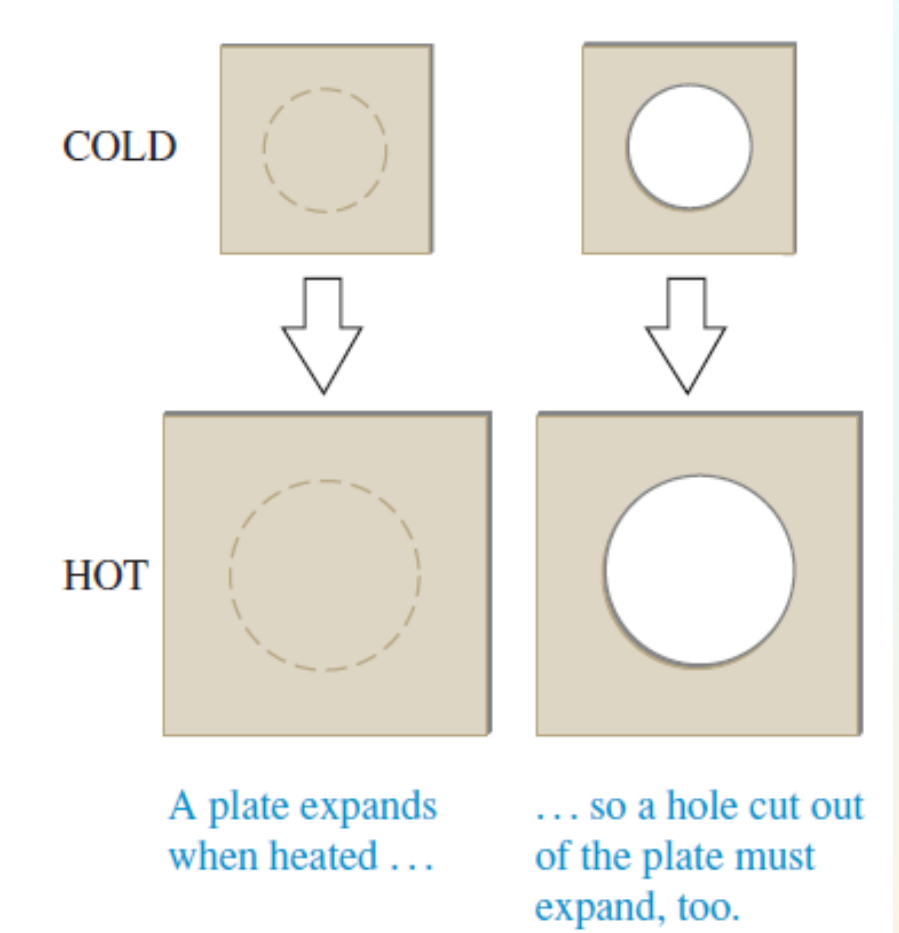


Figure 4. Thermal Expansion in Holes. From. *Sear's & Zemansky's University Physics With Modern Physics*(pp557) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

AREA EXPANSION (SUPERFICIAL EXPANSION)

The formula for area expansion is given as follows:

$$\Delta A = 2\alpha A_o \Delta T$$

$$\Delta A = \gamma A_o \Delta T$$

If a rod has length L_o at temperature T_o, then its length L at a temperature $T = T_o + \Delta T$ is,

$$A = A_o + \Delta A = A_o + \gamma A_o \Delta T = A_o + 2\alpha L_o \Delta T$$

$$\boldsymbol{A = A_o(1 + \gamma \Delta T)}$$

Where: ΔA – change in length (m^2)

A_0– Initial length (m^2)

A – final length (m^2)

ΔT – change in temperature (C^0)

α – coefficient of linear expansion ($/C^0$)

γ - coefficient of superficial expansion ($/C^0$)

$\gamma = 2\alpha$

SAMPLE

CONCEPT CHECK

We cut a circular piece from the rectangular plate. Which ones of the processes given below can help us in passing through the circular piece from the hole?

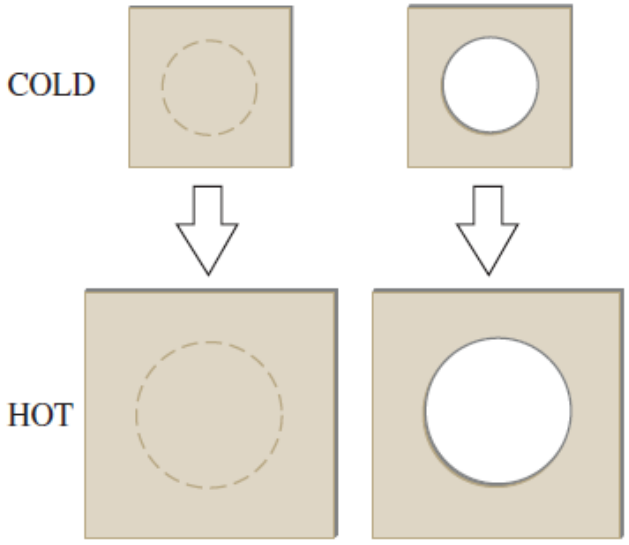


Figure 4. Thermal Expansion in Holes. From. *Sear's & Zemansky's University Physics With Modern Physics*(pp557) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

I. Increasing the temperatures of rectangular plate and circular piece
II. Decreasing the temperature of the circular piece
III. Decreasing the temperatures of the rectangular plate and circular piece

Coefficients of Linear Expansion of Some Materials

Table 17.1 Coefficients of Linear Expansion

Material	$\alpha\ [\mathrm{K}^{-1}$ or $(\mathrm{C}^\circ)^{-1}]$
Aluminum	2.4×10^{-5}
Brass	2.0×10^{-5}
Copper	1.7×10^{-5}
Glass	$0.4–0.9 \times 10^{-5}$
Invar (nickel–iron alloy)	0.09×10^{-5}
Quartz (fused)	0.04×10^{-5}
Steel	1.2×10^{-5}

Table 1 . Coefficients of Linear Expansion of Some Materials . From. *Sear's & Zemansky's University Physics With Modern Physics*(pp557) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

VOLUME EXPANSION

Increasing temperature usually causes increases in volume for both solid and liquid materials.

$$\Delta V = \beta V_o \Delta T$$

Where: ΔV – change in volume
V_0– Initial volume
ΔT – change in temperature
β – coeffecient of volume expansion

The relationship between coefficient of linear expansion and coefficient of volume expansion is:

$$\beta = 3\alpha$$

Coefficients of Volume Expansion of Some Materials

Table 17.2 Coefficients of Volume Expansion

Solids	β [K^{-1} or $(C^\circ)^{-1}$]	Liquids	β [K^{-1} or $(C^\circ)^{-1}$]
Aluminum	7.2×10^{-5}	Ethanol	75×10^{-5}
Brass	6.0×10^{-5}	Carbon disulfide	115×10^{-5}
Copper	5.1×10^{-5}	Glycerin	49×10^{-5}
Glass	$1.2–2.7 \times 10^{-5}$	Mercury	18×10^{-5}
Invar	0.27×10^{-5}		
Quartz (fused)	0.12×10^{-5}		
Steel	3.6×10^{-5}		

Table 2 . Coefficients of Volume Expansion of Some Materials . From. *Sear's & Zemansky's University Physics With Modern Physics*(pp557) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

SAMPLE

2. A glass flask with volume 200cm3 is filled to the brim with mercury at 20 °C. How much mercury overflows when the temperature of the system is raised to 100 °C? (The coefficient of linear expansion of the glass is $0.40x10^{-5}K^{-1}$. The coefficient of volume expansion of mercury is $18x10^{-5}K^{-1}$)

SOLUTION

$$V_{f_{mercury}} - V_{f_{glass}} = V_o\left[(1+\beta\Delta T)_{mercury} - (1+\beta\Delta T)_{glass}\right]$$

(Mercury expands faster than glass)

$$= 200cm^3\left[\left(1+18x10^{-5}/C^0(100^0C - 20^0C)\right)\right.$$
$$\left.-\left(1+3*0.4x10^{-5}/C^0(100^0C - 20^0C)\right)\right]$$
$$= 2.688cm^3$$

ANSWER: 2.688cm³

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

https://www.physicstutorials.org/home/heat-temperature-and-thermal-expansion/thermal-expansion-and-contraction

https://www.jagranjosh.com/general-knowledge/thermal-expansion-of-solids-liquids-and-gases-1458720400-1

College Physics 2

HEAT TRANSFER (THERMAL EQUILIBRIUM)

MPS Department I FEU Institute of Technology

OBJECTIVES

- Define heat, specific heat and calorimetry
- Solve problems related to calorimetry
- State the Zeroth Law of Thermodynamics
- Explain the First Law of Thermodynamics

HEAT

Heat – it is the energy which travels from high temperature to low temperature in a matter. It is also called **thermal or internal energy.**

Energy transfer that takes place solely because of temperature difference is called *heat flow or heat transfer* and energy transferred in this way is called *heat*

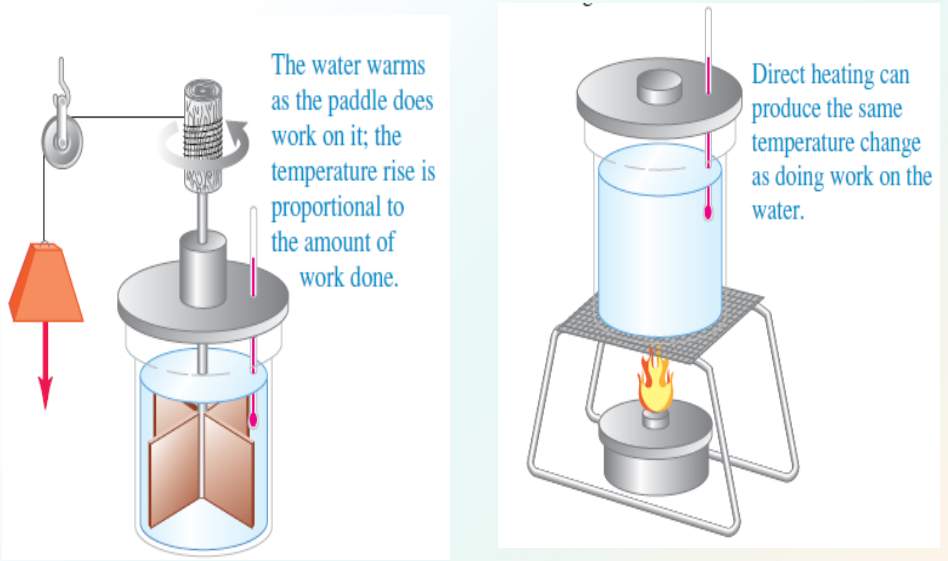


Figure 1. The same temperature change of the same system may be accomplished by (a) doing work on it or (b) adding heat to it. From. *Sear's & Zemansky's University Physics With Modern Physics*(pp562) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

According to Joule, the temperature rise is directly proportional to the amount of work done.

HEAT TRANSFER

Calorimetry means measuring heat.

CONDUCTION

- Transfer of heat energy by molecular and electron collision within a substance (especially solid))
- Materials that transfer heat through this manner is referred to as conductors
- Solids are good conductors because their molecules are far from each other and liquids and gases are poor conductors since their molecules are far from each other

HEAT TRANSFER

CONVECTION

- Transfer of heat energy in a gas or liquid by means of currents in the heated fluid. The fluid moves carrying energy with it
  Example: hot air rising

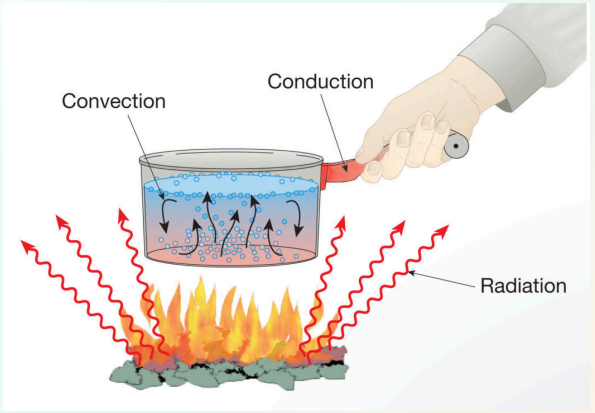
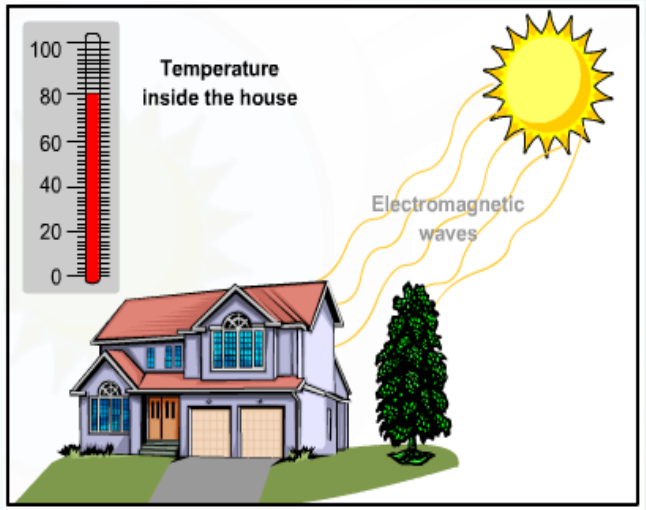


Figure 2. Heat Transfer. From. *https://www.simscale.com/docs/simwiki/heat-transfer-thermal-analysis/what-is-heat-transfer/*

HEAT TRANSFER

RADIATION

- Refers to the transfer of energy at the speed of light through electromagnetic waves
- Radiant Energy- energy that is so radiated in a form of electromagnetic waves
- Dark, rough surfaces absorb more radiant energy and produce heat. Light, smooth materials reflect radiant energy and produce less heat



QUANTITY OF HEAT

Calorie – is defined as the amount of heat required to raise the temperature of 1 gram of water from 14.5°C to 15.5°C.

British Thermal Unit (BTU) – is the quantity of heat required to raise the temperature of 1 pound (weight) of water 1°F

1 cal = 4.186 J
1 kcal = 1000 cal
1 Btu = 778 ft lb = 252 cal

QUANTITY OF HEAT

The quantity of heat Q required to increase the temperature of a mass m of a certain material from T_1 to T_2, is found to be proportional to the change in temperature and the mass of the material.

$$Q = mc\Delta T$$

Where: Q – heat required (J)
m – mass (kg)
c – specific heat (J/kg C^0)
ΔT – change in temperature (C^0)

Specific heat (c) is the amount of heat required to raise the temperature by 1°C for the mass of 1kg.

Specific heat (c) of water:

$$\mathbf{C_{water} = 4186\ J/kg\ C^0 = 1cal/g\ C^0 = 1Btu/lb\ F^0}$$

ZEROTH LAW OF THERMODYNAMICS

If C is initially in thermal equilibrium with both A and B, then A and B are also in thermal equilibrium with each other.

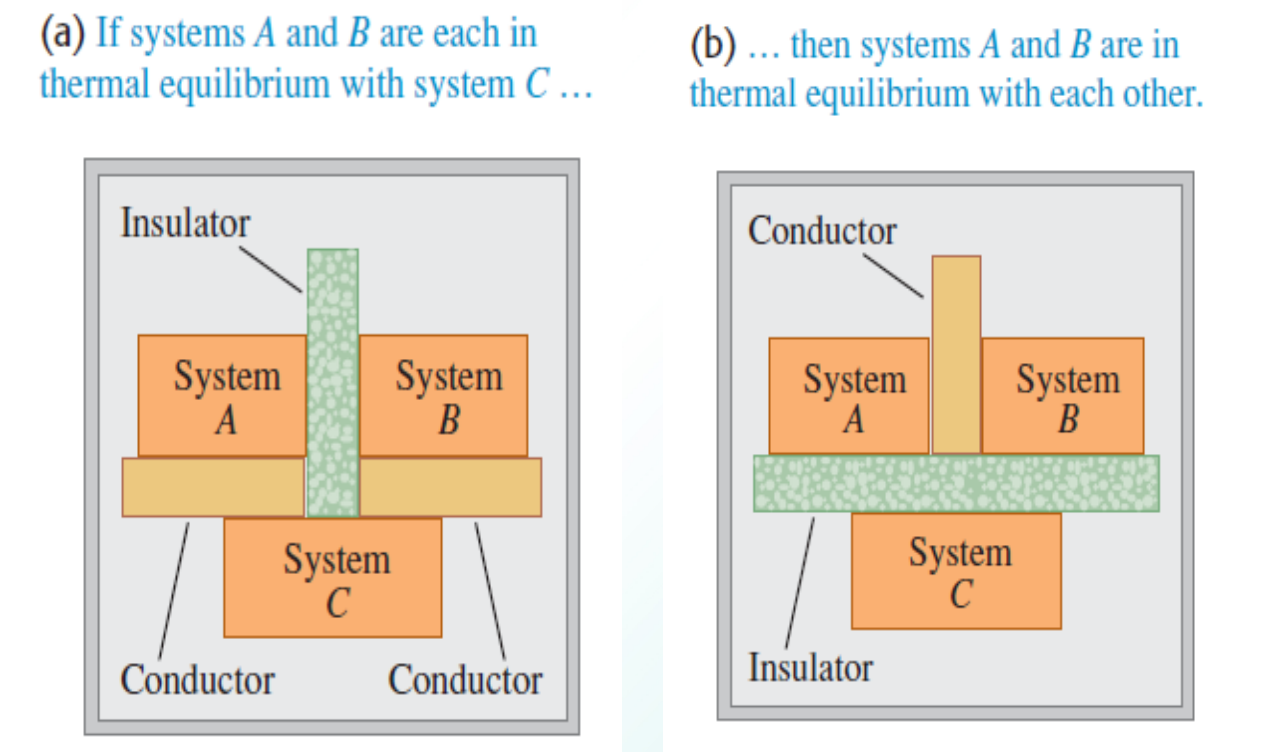


Figure 3. The Zeroth Law of Thermodynamics. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp553) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

ZEROTH LAW OF THERMODYNAMICS

Two systems are in thermal equilibrium if and only if they have the same temperature.

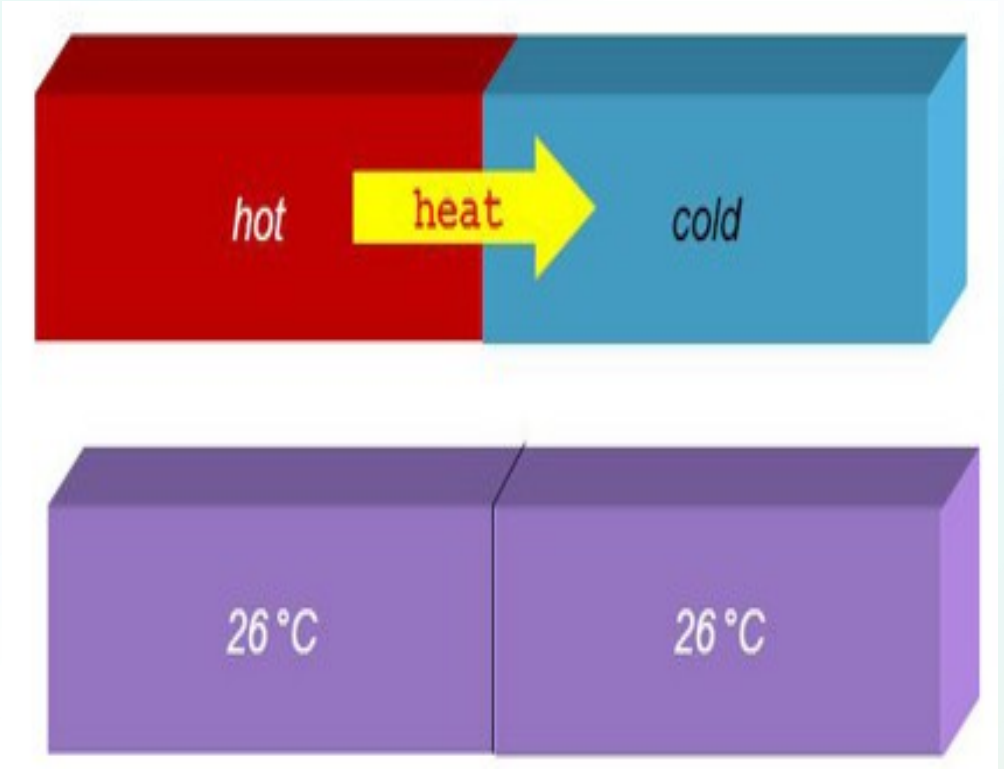


Figure 4. Thermal Equilibrium . Fromhttps://www.assignmentpoint.com/science/physics/thermal-equilibrium.html

CONSERVATION OF ENERGY

Conservation of energy requires that the amount of energy that leaves a body is equal the amount of energy that enters another body in an isolated system.

$$Q_{gain} + Q_{loss} = 0$$

SAMPLE

1. A 80-kg man has ran a fever of 39°C instead of the normal body temperature of 37°C. Assuming that the human body is mostly water, how much heat is required to raise his temperature by that amount? (C_{water} = 4186 J/kg C^0)

SOLUTION

$$Q = mc\Delta T$$
$$= 80\ kg\ (4186\ J/kg\ C^0)\ (2C^{0)})$$
$$= 669{,}760\ J$$

ANSWER: 669.76 kJ

SAMPLE

2. What is the final temperature when 625 grams of water at 75°C loses 7.96×10^4 J?

SOLUTION

$$Q = mc\Delta T$$

$$-7.96 \times 10^4 \text{ J} = 0.625 \text{ kg } (4186 \text{ J/kg } C^0)\ (T_F - 75C^0)$$

$$T_F = 44.6\ ^0C$$

ANSWER: 44.6°C

SAMPLE

3. A 65 g piece of iron at 525°C is put into 635 grams of water at 15°C. What is the final temperature of the water and the iron? (c_{iron} = 0.444 J/g-C)

SOLUTION

$$Q_{gain} + Q_{loss} = 0$$

$$Q_{water} + Q_{iron} = 0$$

$$(mc\Delta T)_{water} + (mc\Delta T)_{iron} = 0$$

$$[0.635kg(4186J/kgC^0)(T_f\text{-}15C^0)] + [0.065kg(444J/kgC^0)(T_f\text{-}525C^0)] = 0$$

$$\mathbf{T_f = 20.5\ ^0C}$$

THERMODYNAMICS

- A branch of science that deals with the properties of matter and their changes due to the effects of heat and work (Shavit and Gutfinger, 2009).

- Comes from the Greek word which means the power of heat (Shavit and Gutfinger, 2009).

- is the study of heat and work

Thermodynamic Properties

- Temperature – degree of hotness
- Pressure – force per unit area
- Density – mass per unit volume

ZEROTH LAW OF THERMODYNAMICS

If *C* is initially is in thermal equilibrium with both *A* and *B*, then *A* and *B* are also in thermal equilibrium with each other.

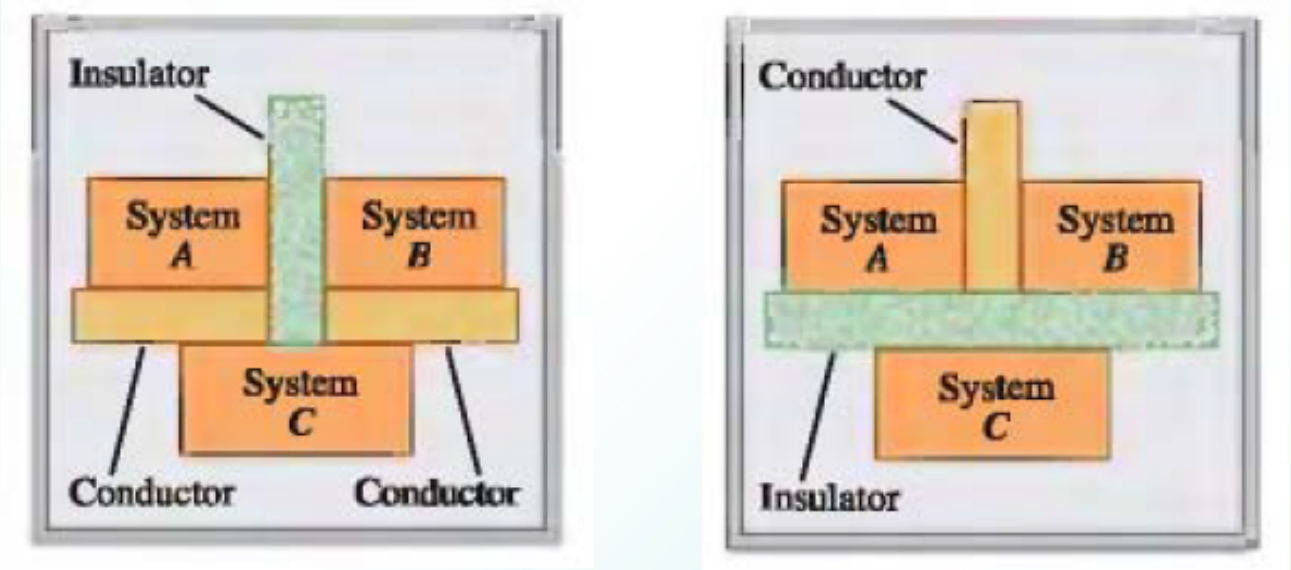


Figure 5 The Zeroth Law of Thermodynamics. From. *Sear's & Zemansky's University Physics With Modern Physics* (pp553) by Young, H. D., Freedman, R. A. & A. Lewis Ford(2016). England: Pearson.

QUANTITY OF HEAT

PHASE CHANGE

results when heat is absorbed or released without a change in temperature

Melting/Freezing

To calculate for the amount of heat associated with a change in the object's phase, we use:

$$Q = \pm mL_f$$

- Q: heat
- m: mass
- L_f: latent heat of fusion

amount of heat required to change a substance from solid to liquid

Use $+L_f$ for melting and $-L_f$ for freezing

QUANTITY OF HEAT

PHASE CHANGE

results when heat is absorbed or released without a change in temperature

Melting/Freezing

To calculate for the amount of heat associated with a change in the object's phase, we use:

$$Q = \pm mL_V$$

- Q: heat
- m: mass
- L_V: latent heat of vaporization

amount of heat required to change a substance from liquid to gas

Use $+L_v$ for melting and $-L_v$ for freezing

PHASE CHANGE

A plot of temperature versus energy added when 1.00 g of ice initially at 30.0°C is converted to steam at 120.0°C.

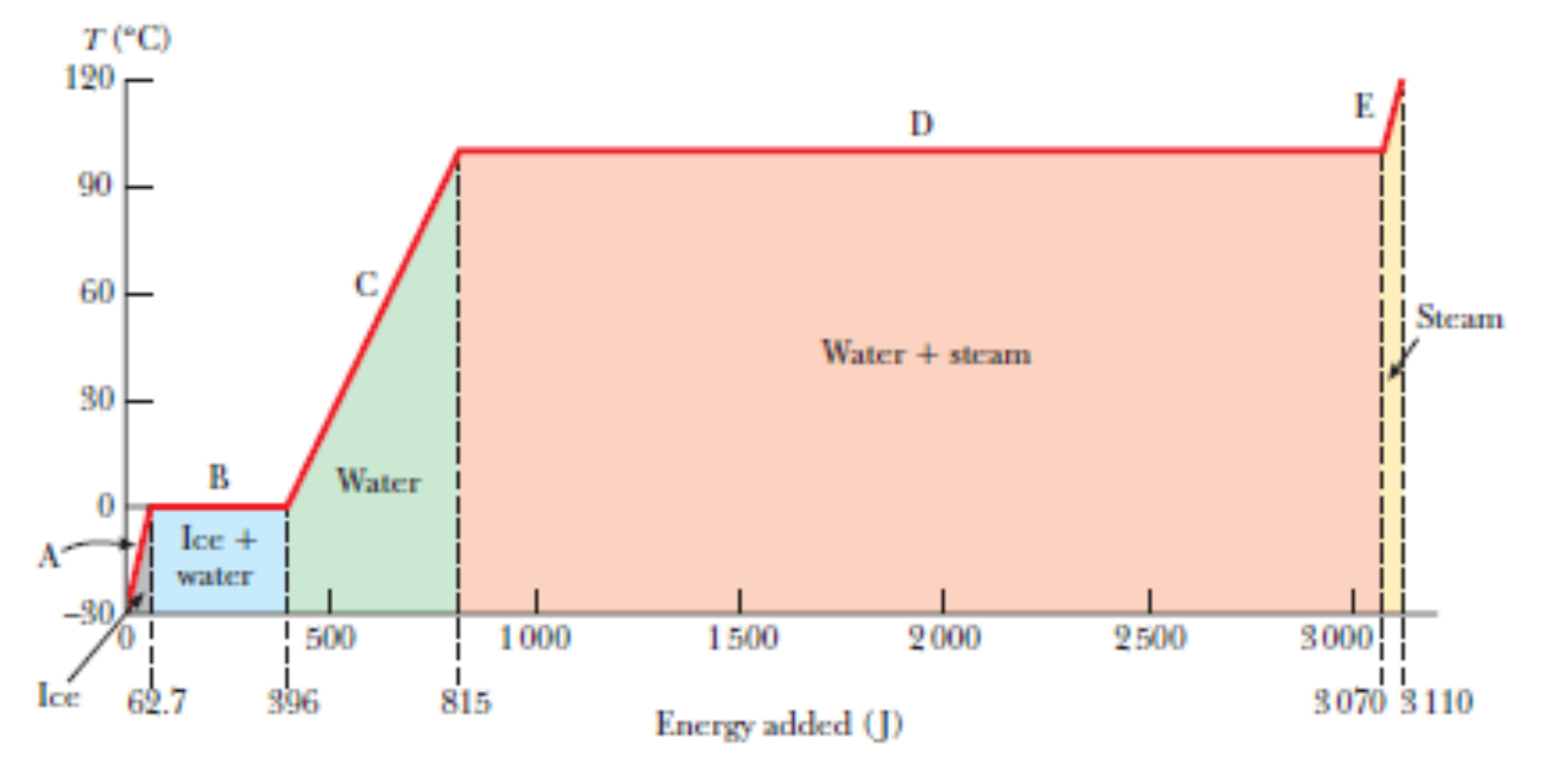


Figure 6 Plot of Temperature vs Energy. From. *Physics for Scientists and Engineers with Modern Physics*(pp 599) by Serway, R. A., & Jewett, J. W. (2019). Australia: Cengage Learning.

Heats of Fusion and Vaporization

Substance	Normal Melting Point		Heat of Fusion, L_f (J/kg)	Normal Boiling Point		Heat of Vaporization, L_v (J/kg)
	K	°C		K	°C	
Helium	*	*	*	4.216	−268.93	20.9×10^3
Hydrogen	13.84	−259.31	58.6×10^3	20.26	−252.89	452×10^3
Nitrogen	63.18	−209.97	25.5×10^3	77.34	−195.8	201×10^3
Oxygen	54.36	−218.79	13.8×10^3	90.18	−183.0	213×10^3
Ethanol	159	−114	104.2×10^3	351	78	854×10^3
Mercury	234	−39	11.8×10^3	630	357	272×10^3
Water	273.15	0.00	334×10^3	373.15	100.00	2256×10^3
Sulfur	392	119	38.1×10^3	717.75	444.60	326×10^3
Lead	600.5	327.3	24.5×10^3	2023	1750	871×10^3
Antimony	903.65	630.50	165×10^3	1713	1440	561×10^3
Silver	1233.95	960.80	88.3×10^3	2466	2193	2336×10^3
Gold	1336.15	1063.00	64.5×10^3	2933	2660	1578×10^3
Copper	1356	1083	134×10^3	1460	1187	5069×10^3

*A pressure in excess of 25 atmospheres is required to make helium solidify. At 1 atmosphere pressure, helium remains a liquid down to absolute zero.

SAMPLE PROBLEMS

Calculate the total energy needed for a 100 g piece of ice at -40°C to convert it into steam of 110°C.

SOLUTION

$$Q_{total} = mc_{ice}\Delta T + mL_f + mc_{water}\Delta T + mL_v + mc_{vapor}\Delta T$$

$$= (0.1kg)\left(2090\,{}^{J}/_{kgC^0}\right)[(0C^0) - (-40C^0)] +$$

$$(0.1kg)\left(3.35x10^5\,{}^{J}/_{kg}\right) +$$

$$(0.1kg)\left(4186\,{}^{J}/_{kgC^0}\right)[(100C^0) - (0C^0)] +$$

$$(0.1kg)\left(2.26x10^6\,{}^{J}/_{kg}\right) +$$

$$(0.1kg)\left(2110\,{}^{J}/_{kgC^0}\right)[(110C^0) - (100C^0)]$$

$$= \mathbf{3.12x10^5 J}$$

FIRST LAW OF THERMODYNAMICS

The change in the **internal energy** ΔU of a **closed system** is equal to the **energy added to the system through heat** Q minus the **work done by the system on its surroundings** W.

$$\Delta U = Q - W$$

Closed System – a system that can exchange energy, but not matter, with its surroundings

Internal Energy – the sum of the kinetic energies of all its constituent particles plus the sum of all the potential energies of the interaction among the particles.

Sign Conventions
+Q – heat added
-Q – heat loss
+W – by the system
-W – on the system

FIRST LAW OF THERMODYNAMICS

Let's Try!

Tell whether the change in internal energy is due to heat (Q) or work (W):

1. The soup in a pan sitting on an electric stove gets hot.
2. Your hands are warmed when they're placed near fire.
3. The atmosphere warms a re-entering spacecraft.

SAMPLE PROBLEMS

1. 3000 J of heat is added to a system and 2500 J of work is done by the system. What is the change in its internal energy?

Solution:
The equation of the first law of thermodynamics

$\Delta U = Q - W$
$\Delta U = 3000 - 2500$
$\Delta U = 500$ Joules
Internal energy increases by 500 Joules.

The sign conventions :
Q is positive if the heat added to the system
W is positive if work is done by the system
Q is negative if heat leaves the system
W is negative if work is done on the system

SAMPLE PROBLEMS

2. Suppose there is a heat transfer of 40.00 J to a system, while the system does 10.00 J of work. Later, there is a heat transfer of 25.00 J out of the system while 4.00 J of work is done on the system. What is the net change in internal energy of the system?

Solution:

The net heat transfer is the heat transfer into the system minus the heat transfer out of the system, or

$$Q = 40.00\ \text{J} - 25.00\ \text{J} = 15.00\ \text{J}.$$

Similarly, the total work is the work done by the system minus the work done on the system, or

$$W = 10.00\ \text{J} - 4.00\ \text{J} = 6.00\ \text{J}.$$

Thus the change in internal energy is given by the first law of thermodynamics:

$$\Delta U = Q - W = 15.00\ \text{J} - 6.00\ \text{J} = 9.00\ \text{J}.$$

REFERENCES

Serway, R. A., & Jewett, J. W. (2019). Physics for Scientists and Engineers with Modern Physics Tenth Edition. Australia: Cengage Learning.

Young, H. D., & Freedman, R. A. (2016). Sear's & Zemansky's University Physics With Modern Physics. England: Pearson.

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